The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Proto-Germ-Generating Substrate Pre-Proto-Germ
Observer-Relevant Core Collapse Theorem

Abstract

The current active primitive-reservoir observer-reduction frontier already moved the live burden below the proto-germ package: the present proto-germ package was derived from a more primitive proto-germ-generating substrate pre-proto-germ dynamics package. Under the recorded routing safeguard and the benchmark one-metric upstream compression rule, however, the honest next move at that level is not another same-output deeper-origin relay that leaves the observer-relevant pre-proto-germ data unchanged. The sharper benchmark-facing gain is to prove a genuinely smaller theorem-level collapse strictly inside the observer-relevant pre-proto-germ package itself.

The present note takes that route. It isolates the *observer-relevant pre-proto-germ core*: the pre-proto-germ state space and compact body, the projection to the recorded proto-germ body together with its proto section, the pre-proto-germ restriction section and exact shell, the embedded zero core and split, the hidden charge, the observer-compatible exact zero-response realization, and the zero-core solvability / coercive package. The lower minimizer hierarchy from the pre-proto-germ package to the fixed germ, seed, foundation, ground, origin, precursor, lift, and substrate fibers is then shown not to be an independent benchmark-facing burden. Any benchmark-faithful completion of the same observer-relevant core is benchmark-equivalent, and a canonical completion is obtained by composing the pre-proto proto minimizer with the already fixed downstream proto-germ lower minimizers.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. The positive result is therefore narrower than a completed first-principles derivation and sharper than another same-output relay: the live pre-proto-germ burden now collapses to a smaller observer-relevant core. What remains open is to derive or justify that smaller core from still more primitive substrate structure, or else prove a still sharper collapse strictly inside it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is now constrained by two benchmark-facing routing facts. First, the current proto-germ-frontier theorem already removed the proto-germ package as the primitive stopping point by deriving it from a more primitive pre-proto-germ-level substrate package. Second, the repository has already fixed a safeguard against recursive depth drift: once an observer-relevant core is fixed, another deeper-origin relay that preserves that same core is no longer honest benchmark-facing progress merely by being deeper.

That safeguard matters here. The next honest move below the current pre-proto-germ frontier is therefore not to spend a manuscript on another same-output relay unless that relay genuinely changes the observer-relevant burden. The sharper default at current scope is instead to ask whether the full recorded pre-proto-germ package contains internal freedom that is benchmark-neutral and can be collapsed away without reopening the benchmark exact-closure package.

The present note answers that question affirmatively. Inside the recorded proto-germ-generating substrate pre-proto-germ dynamics package, the lower minimizer ladder down to the fixed germ, seed, foundation, ground, origin, precursor, lift, and substrate fibers is not an independent observer-relevant burden. Once the pre-proto-germ package is restricted to the data that actually control the projected exact-shell transport to the current proto-germ package and the fixed benchmark outputs, that omitted lower ladder can be supplied canonically and any alternative supply of the same omitted data is benchmark-equivalent.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not reopen older screened layers. Its positive role is narrower and benchmark facing: compress the live pre-proto-germ burden to the part of the package that still matters observer-relevantly.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded downstream ladder and proto-germ target)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes:*

1. *the lower recorded fibers and targets*

$$\mathcal{B}^{\mathrm{germ}}_{\sigma}, \qquad \mathcal{B}^{\mathrm{seed}}_{\sigma}, \qquad \mathcal{F}^{\mathrm{fdn}}_{\sigma}, \qquad \mathcal{G}^{\mathrm{grd}}_{\sigma}, \qquad \mathcal{O}^{\mathrm{orig}}_{\sigma},$$

$$\mathcal{P}^{\mathrm{prec}}_{\sigma}, \qquad \mathcal{L}^{\mathrm{lift}}_{\sigma}, \qquad \mathcal{M}^{\mathrm{sub}}_{\sigma}, \qquad E^{\mathrm{sub}}_{\sigma}, \qquad Q^{\mathrm{eqn}}_{\sigma},$$

together with the positive-definite pairing

$$\mathfrak{s}^{\mathrm{sub}}_{\sigma} : E^{\mathrm{sub}}_{\sigma} \times E^{\mathrm{sub}}_{\sigma} \to \mathbb{R};$$

2. *the hidden fiber space* $\mathcal{H}^{\mathrm{hid}}_{\sigma}$ *with compact hidden body* $D^{\mathrm{hid}}_{\sigma} \subset \mathcal{H}^{\mathrm{hid}}_{\sigma}$*;*

3. *the recorded proto-germ package* $\mathfrak{P}^{\mathrm{proto}}$*, namely a finite-dimensional state space* $\mathcal{U}^{\mathrm{proto}}_{\sigma}$*, a compact convex body* $\mathcal{B}^{\mathrm{proto}}_{\sigma} \subset \mathcal{U}^{\mathrm{proto}}_{\sigma}$*, smooth lower minimizers*

$$\mathfrak{f}^{\mathrm{proto,germ}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{proto,seed}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{proto,fdn}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{proto,grd}}_{\sigma},$$

$$\mathfrak{f}^{\mathrm{proto,orig}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{proto,prec}}_{\sigma}, \quad \mathfrak{f}^{\mathrm{proto,lift}}_{\sigma}, \quad \mathfrak{m}^{\mathrm{proto}}_{\sigma},$$

a smooth proto-germ restriction section

$$\Gamma^{\mathrm{proto}}_{\sigma} : \mathcal{U}^{\mathrm{proto}}_{\sigma} \to E^{\mathrm{sub}}_{\sigma},$$

the exact proto-germ shell

$$\mathcal{Z}^{\mathrm{proto}}_{\sigma} \equiv \left(\Gamma^{\mathrm{proto}}_{\sigma}\right)^{-1}(0) \cap \mathcal{B}^{\mathrm{proto}}_{\sigma},$$

a closed embedded zero core $\mathcal{Z}^{0,\mathrm{proto}}_{\sigma} \subset \mathcal{Z}^{\mathrm{proto}}_{\sigma}$*, a split diffeomorphism*

$$\Phi^{\mathrm{proto}}_{\sigma} : \mathcal{Z}^{0,\mathrm{proto}}_{\sigma} \times D^{\mathrm{hid}}_{\sigma} \xrightarrow{\cong} \mathcal{Z}^{\mathrm{proto}}_{\sigma},$$

a hidden charge

$$\mathcal{Q}^{\mathrm{proto}}_{\sigma} : \mathcal{H}^{\mathrm{hid}}_{\sigma} \to Q^{\mathrm{eqn}}_{\sigma},$$

an observer-compatible exact zero-response realization

$$\zeta^{\mathrm{proto}}_{\sigma} : \mathcal{I}^{\mathrm{for}}_{\sigma} \hookrightarrow \mathcal{Z}^{0,\mathrm{proto}}_{\sigma},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}^{0,\mathrm{proto}}_{\sigma}$*, recorded through* $\widetilde{K}^{0,\mathrm{proto}}_{\sigma}$ *and* $P^{\mathrm{proto,co}}_{\sigma}$.

Remark 1. *Definitions 1 and 2 are fixed inputs. The one operative metric, the ordinary matter route, and the recorded proto-germ package are not reopened here. The present note asks only which part of the pre-proto-germ package still carries independent benchmark-facing burden once those fixed downstream data are already in hand.*

3. Full Pre-Proto-Germ Package

Definition 3 (Proto-germ-generating substrate pre-proto-germ dynamics)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A proto-germ-generating substrate pre-proto-germ dynamics package* $\mathfrak{Q}^{\mathrm{preproto}}$ *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean pre-proto-germ state space* $\mathcal{V}_\sigma^{\mathrm{preproto}}$*, a compact convex pre-proto-germ body*

$$\mathcal{B}_\sigma^{\mathrm{preproto}} \subset \mathcal{V}_\sigma^{\mathrm{preproto}}$$

with nonempty interior, an affine surjection

$$\Pi_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to \mathcal{U}_\sigma^{\mathrm{proto}},$$

and smooth pre-proto-germ fiber minimizers

$$\mathfrak{f}_\sigma^{\mathrm{preproto,proto}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,germ}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}},$$

$$\mathfrak{f}_\sigma^{\mathrm{preproto,grd}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{preproto}},$$

satisfying

$$\Pi_\sigma^{\mathrm{preproto}}(\mathcal{B}_\sigma^{\mathrm{preproto}}) = \mathcal{B}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}},$$

together with the transport identities

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,germ}} = \mathfrak{f}_\sigma^{\mathrm{proto,germ}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,seed}} = \mathfrak{f}_\sigma^{\mathrm{proto,seed}},$$

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}} = \mathfrak{f}_\sigma^{\mathrm{proto,fdn}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,grd}} = \mathfrak{f}_\sigma^{\mathrm{proto,grd}},$$

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,orig}} = \mathfrak{f}_\sigma^{\mathrm{proto,orig}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,prec}} = \mathfrak{f}_\sigma^{\mathrm{proto,prec}},$$

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,lift}} = \mathfrak{f}_\sigma^{\mathrm{proto,lift}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{m}_\sigma^{\mathrm{preproto}} = \mathfrak{m}_\sigma^{\mathrm{proto}}.$$

2. *A smooth pre-proto-germ restriction section*

$$\Gamma_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to E_\sigma^{\mathrm{sub}},$$

the pre-proto-germ functional

$$\mathscr{W}_\sigma^{\mathrm{preproto}}(q) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}(\Gamma_\sigma^{\mathrm{preproto}}(q), \Gamma_\sigma^{\mathrm{preproto}}(q)),$$

and the exact pre-proto-germ shell

$$\mathcal{Z}_\sigma^{\mathrm{preproto}} \equiv (\Gamma_\sigma^{\mathrm{preproto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{preproto}},$$

satisfying

$$\Gamma_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \Gamma_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} : \mathcal{Z}_\sigma^{\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}.$$

3. *A closed embedded pre-proto-germ zero core*

$$\mathcal{Z}_\sigma^{0,\mathrm{preproto}} \subset \mathcal{Z}_\sigma^{\mathrm{preproto}},$$

a diffeomorphism

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{0,\mathrm{proto}},$$

and a split diffeomorphism

$$\Phi_\sigma^{\mathrm{preproto}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}}$$

obeying the projected split identity

$$\Phi_\sigma^{\mathrm{proto}}(z,\eta) = \Pi_\sigma^{\mathrm{preproto}}\Big(\Phi_\sigma^{\mathrm{preproto}}\big((\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}})^{-1}(z),\eta\big)\Big)$$

for every $z \in \mathcal{Z}_\sigma^{0,\mathrm{proto}}$ *and every* $\eta \in D_\sigma^{\mathrm{hid}}$.

4. *A hidden charge*

$$\mathcal{Q}_\sigma^{\mathrm{preproto}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$

a smooth observer-compatible exact pre-proto-germ zero-response realization

$$\zeta_\sigma^{\mathrm{preproto}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{preproto}},$$

satisfying

$$\mathcal{Q}_\sigma^{\mathrm{proto}} = \mathcal{Q}_\sigma^{\mathrm{preproto}}, \qquad \zeta_\sigma^{\mathrm{proto}} = \Pi_\sigma^{\mathrm{preproto}} \circ \zeta_\sigma^{\mathrm{preproto}},$$

and a fixed-data solvability / coercive package on $\mathcal{Z}_\sigma^{0,\mathrm{preproto}}$*, recorded through* $\widetilde{K}_\sigma^{0,\mathrm{preproto}}$ *and* $P_\sigma^{\mathrm{preproto,co}}$*, whose projection through*

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}}$$

recovers the recorded proto-germ kernel chart $\widetilde{K}_\sigma^{0,\mathrm{proto}}$ *and orthogonal projector* $P_\sigma^{\mathrm{proto,co}}$.

The package is benchmark faithful *if it preserves the benchmark outputs* (1)*, introduces no second operative metric, introduces no enlarged low-energy matter law, and is presented as a deeper substrate-side source for the current proto-germ package rather than as a replacement benchmark theory.*

Remark 2. *Definition 3 is the current full pre-proto-germ package. The present note does not go one layer deeper below it. Instead, it asks whether every datum listed there still needs to be treated as an independent live burden once the current proto-germ target and the fixed benchmark outputs are held fixed.*

4. Observer-Relevant Pre-Proto-Germ Core

Definition 4 (Observer-relevant pre-proto-germ core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The* observer-relevant pre-proto-germ core $\mathfrak{Q}^{\mathrm{core}}$ *for sector* σ *consists of the following data:*

1. *the pre-proto-germ state space* $\mathcal{V}_\sigma^{\mathrm{preproto}}$*, compact convex body* $\mathcal{B}_\sigma^{\mathrm{preproto}} \subset \mathcal{V}_\sigma^{\mathrm{preproto}}$*, projection*

$$\Pi_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to \mathcal{U}_\sigma^{\mathrm{proto}},$$

   and proto section

$$\mathfrak{f}_\sigma^{\mathrm{preproto,proto}} : \mathcal{B}_\sigma^{\mathrm{proto}} \to \mathcal{B}_\sigma^{\mathrm{preproto}}$$

   satisfying

$$\Pi_\sigma^{\mathrm{preproto}}(\mathcal{B}_\sigma^{\mathrm{preproto}}) = \mathcal{B}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}};$$

2. *the restriction section*

$$\Gamma_\sigma^{\mathrm{preproto}} : \mathcal{V}_\sigma^{\mathrm{preproto}} \to E_\sigma^{\mathrm{sub}},$$

   the induced functional

$$\mathscr{W}_\sigma^{\mathrm{preproto}}(q) \equiv \mathfrak{s}_\sigma^{\mathrm{sub}}(\Gamma_\sigma^{\mathrm{preproto}}(q), \Gamma_\sigma^{\mathrm{preproto}}(q)),$$

   and the exact shell

$$\mathcal{Z}_\sigma^{\mathrm{preproto}} \equiv (\Gamma_\sigma^{\mathrm{preproto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{preproto}},$$

   with

$$\Gamma_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \Gamma_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{preproto}}} : \mathcal{Z}_\sigma^{\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}};$$

3. *the embedded zero core* $\mathcal{Z}_\sigma^{0,\mathrm{preproto}} \subset \mathcal{Z}_\sigma^{\mathrm{preproto}}$*, the zero-core diffeomorphism*

$$\Pi_\sigma^{\mathrm{preproto}}\big|_{\mathcal{Z}_\sigma^{0,\mathrm{preproto}}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{0,\mathrm{proto}},$$

   and the split

$$\Phi_\sigma^{\mathrm{preproto}} : \mathcal{Z}_\sigma^{0,\mathrm{preproto}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{preproto}}$$

   satisfying the projected split identity of Definition 3(3);

4. *the hidden charge* $\mathcal{Q}_\sigma^{\mathrm{preproto}}$*, the observer-compatible exact zero-response realization* $\zeta_\sigma^{\mathrm{preproto}}$*, and the fixed-data solvability / coercive package* $\widetilde{K}_\sigma^{0,\mathrm{preproto}}, P_\sigma^{\mathrm{preproto,co}}$ *satisfying the projected identities of Definition 3(4);*

5. *benchmark faithfulness in the sense of Definition 3.*

By definition, the omitted lower minimizer ladder

$$\mathfrak{f}_\sigma^{\mathrm{preproto,germ}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,grd}},$$
$$\mathfrak{f}_\sigma^{\mathrm{preproto,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{preproto}}$$

is not *part of the core.*

Definition 5 (Completion of the observer-relevant core)**.** *A* benchmark-faithful completion *of* $\mathfrak{Q}^{\mathrm{core}}$ *is a full pre-proto-germ package in the sense of Definition 3 whose data agree with the given core and that adds only the omitted lower minimizer ladder. Two benchmark-faithful completions are* core-equivalent *if they agree on every datum of Definition 4.*

Remark 3. *Definition 4 isolates the pre-proto-germ data that control projected exact-shell transport to the fixed proto-germ package and the fixed benchmark outputs. The omitted lower minimizer ladder still matters as recorded package structure, but the present question is whether it remains an independent benchmark-facing burden.*

5. Canonical Completion and Core Compression

Proposition 1 (Canonical completion of the omitted lower ladder)**.** *Fix an observer-relevant pre-proto-germ core* $\mathfrak{Q}^{\mathrm{core}}$*. Then there is a canonical benchmark-faithful completion obtained by*

$$\widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,germ}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,germ}}, \qquad \widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,seed}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,seed}},$$

$$\widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,fdn}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,fdn}}, \qquad \widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,grd}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,grd}},$$

$$\widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,orig}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,orig}}, \qquad \widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,prec}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,prec}},$$

$$\widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,lift}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{f}_\sigma^{\mathrm{proto,lift}}, \qquad \widehat{\mathfrak{m}_\sigma^{\mathrm{preproto}}} \equiv \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} \circ \mathfrak{m}_\sigma^{\mathrm{proto}}.$$

Proof. Each displayed composition has the correct codomain $\mathcal{B}_\sigma^{\mathrm{preproto}}$ because $\mathfrak{f}_\sigma^{\mathrm{preproto,proto}}$ maps $\mathcal{B}_\sigma^{\mathrm{proto}}$ into $\mathcal{B}_\sigma^{\mathrm{preproto}}$. Using

$$\Pi_\sigma^{\mathrm{preproto}} \circ \mathfrak{f}_\sigma^{\mathrm{preproto,proto}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{proto}}}$$

from Definition 4(1), we obtain immediately

$$\Pi_\sigma^{\mathrm{preproto}} \circ \widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,germ}}} = \mathfrak{f}_\sigma^{\mathrm{proto,germ}}, \qquad \Pi_\sigma^{\mathrm{preproto}} \circ \widehat{\mathfrak{f}_\sigma^{\mathrm{preproto,seed}}} = \mathfrak{f}_\sigma^{\mathrm{proto,seed}},$$

and likewise for the foundation, ground, origin, precursor, lift, and substrate maps. All other data required by Definition 3 are already part of the core. Therefore the displayed compositions provide a benchmark-faithful completion. □

Definition 6 (Benchmark-equivalent completions)**.** *Two benchmark-faithful completions of the same observer-relevant core are* benchmark-equivalent *if they induce:*

1. *the same projected exact shell* $\mathcal{Z}_\sigma^{\mathrm{proto}}$*,*
2. *the same projected zero core* $\mathcal{Z}_\sigma^{0,\mathrm{proto}}$*,*
3. *the same projected split* $\Phi_\sigma^{\mathrm{proto}}$*,*
4. *the same projected hidden charge* $\mathcal{Q}_\sigma^{\mathrm{proto}}$*,*
5. *the same projected exact zero-response realization* $\zeta_\sigma^{\mathrm{proto}}$*,*
6. *the same projected fixed-data solvability / coercive package* $\widetilde{K}_\sigma^{0,\mathrm{proto}}, P_\sigma^{\mathrm{proto,co}}$*,*
7. *and the same benchmark outputs* (1)*.*

Proposition 2 (Core-equivalent completions are benchmark-equivalent)**.** *Any two core-equivalent benchmark-faithful completions are benchmark-equivalent in the sense of Definition 6.*

Proof. By definition, core-equivalent completions agree on the projection $\Pi_\sigma^{\mathrm{preproto}}$, the proto section $\mathfrak{f}_\sigma^{\mathrm{preproto,proto}}$, the restriction section $\Gamma_\sigma^{\mathrm{preproto}}$, the exact shell $\mathcal{Z}_\sigma^{\mathrm{preproto}}$, the zero core $\mathcal{Z}_\sigma^{0,\mathrm{preproto}}$, the split $\Phi_\sigma^{\mathrm{preproto}}$, the hidden charge $\mathcal{Q}_\sigma^{\mathrm{preproto}}$, the exact zero-response realization $\zeta_\sigma^{\mathrm{preproto}}$, the kernel chart $\widetilde{K}_\sigma^{0,\mathrm{preproto}}$, the orthogonal projector $P_\sigma^{\mathrm{preproto,co}}$, and the benchmark-faithfulness condition. Those are exactly the data that determine the seven items listed in Definition 6. The omitted lower minimizer ladder is not used in any of those determinations. Therefore every pair of core-equivalent completions is benchmark-equivalent. □

Proposition 3 (The live proto-germ frontier depends only on the observer-relevant core)**.** *Fix an observer-relevant pre-proto-germ core. Then every benchmark-faithful completion of that core induces the same observer-relevant proto-germ frontier data and therefore the same downstream benchmark-facing consequences already recorded on the active primitive-reservoir line.*

Proof. By Proposition 2, every benchmark-faithful completion of the fixed core induces the same projected exact shell, zero core, split, hidden charge, exact zero-response realization, fixed-data kernel package, and benchmark outputs. These are precisely the pre-proto-to-proto transport data that matter observer-relevantly for the current proto-germ frontier and the fixed benchmark one-metric package. Since the omitted lower minimizer ladder does not alter those data, it does not alter the observer-relevant proto-germ frontier or any downstream benchmark-facing consequence already recorded from that frontier. □

6. Main Theorem

Theorem 1 (Observer-relevant pre-proto-germ core collapse)**.** *Fix the primitive continuation data of Definition 1, the recorded downstream ladder and proto-germ target of Definition 2, and a benchmark-faithful full pre-proto-germ package in the sense of Definition 3. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the live benchmark-facing burden inside that full pre-proto-germ package collapses to the observer-relevant core of Definition 4. More precisely:*

1. *the omitted lower minimizer ladder*

$$\mathfrak{f}_\sigma^{\mathrm{preproto,germ}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,seed}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,fdn}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,grd}},$$
$$\mathfrak{f}_\sigma^{\mathrm{preproto,orig}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,prec}}, \quad \mathfrak{f}_\sigma^{\mathrm{preproto,lift}}, \quad \mathfrak{m}_\sigma^{\mathrm{preproto}}$$

   is not an independent benchmark-facing burden at current scope;

2. *every observer-relevant pre-proto-germ core admits a canonical benchmark-faithful completion by Proposition 1;*

3. *any two benchmark-faithful completions of the same observer-relevant core are benchmark-equivalent and induce the same current proto-germ-frontier data by Propositions 2 and 3;*

4. *the benchmark one-metric package remains fixed: the operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged;*

5. *consequently the remaining honest burden below the current proto-germ-frontier theorem is no longer the full pre-proto-germ package as such, and still less another same-output deeper-origin relay that leaves this core unchanged. What remains is to derive or justify the observer-relevant pre-proto-germ core itself from still more primitive substrate structure, or else prove a still sharper collapse strictly inside that core.*

Proof. Item (1) is the content of Propositions 2 and 3: once the observer-relevant core is fixed, varying the omitted lower minimizer ladder does not change any benchmark-facing output. Item (2) is Proposition 1. Item (3) is the combination of Propositions 2 and 3. Item (4) is part of benchmark faithfulness and the fixed outputs (1). Item (5) is the routing consequence of the previous items together with the benchmark compression safeguard already fixed on the active line. □

Corollary 1 (What remains open after the core collapse)**.** *After Theorem 1, the remaining honest burden is no longer to treat the full pre-proto-germ lower minimizer ladder as primitive benchmark-facing data. That burden has collapsed to the observer-relevant pre-proto-germ core. What remains open is why that smaller core should hold, or whether an even sharper theorem can collapse that core further without changing the same benchmark one-metric package and claim boundary.*

Proof. Theorem 1 shows that the omitted lower ladder is benchmark-neutral once the observer-relevant core is fixed. Therefore the next honest open burden lies at the level of that core or below it, not at the level of another same-output relay or another full-package restatement. □

7. Conclusion

The positive result of this note is narrower than a completed first-principles derivation and sharper than another same-output pre-proto-germ relay. The current frontier had already moved the active burden below the proto-germ package by deriving that package from a more primitive pre-proto-germ class. The present theorem then shows that, inside that current pre-proto-germ class itself, the live benchmark-facing burden is smaller than the full recorded package: the omitted lower minimizer ladder is not an independent observer-relevant burden once the pre-proto-germ core is fixed.

The scope remains disciplined. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to completed derivation, and nothing here licenses stronger promotion language.

The open burden is therefore cleaner. The next benchmark-facing manuscript should derive or justify the observer-relevant pre-proto-germ core from still more primitive substrate structure, unless the sharper honest gain available instead is a further theorem-level collapse strictly inside that core. Until then, the present note should be read for what it is: a genuine internal compression of the active pre-proto-germ frontier in Æther-flow Interpretation of Relativity, not a claim that the first-principles program is finished.